Here's a **step-by-step lab guide** to:

1. **Create IAM Roles from the AWS GUI**
2. **Attach the Role to an EC2 instance**
3. **Install AWS CLI on the EC2**
4. **Use the CLI to control AWS resources**

🧪 Lab: Use IAM Role & AWS CLI on EC2 to Manage AWS Resources

✅ Part 1: Create IAM Role from AWS Console (GUI)

1. Go to **IAM > Roles** → Click **"Create Role"**

- **Trusted Entity Type**: Choose **AWS service**
- **Use case**: Select **EC2**
- Click **Next**

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2. Attach Permissions Policies

Choose a policy (or more) depending on what resources the EC2 instance should manage.

📌 Example: Give access to S3
- Select: `AmazonS3FullAccess`
(For read-only: `AmazonS3ReadOnlyAccess`)

Click **Next**

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3. Name the Role

- Role Name: `EC2S3AccessRole`
- Optional: Add description and tags

Click **Create Role**

✅ Part 2: Attach IAM Role to EC2 Instance

1. Go to **EC2 > Instances**
- Select the running instance
- Click **Actions > Security > Modify IAM Role**

2. Attach Role
- Select `EC2S3AccessRole` from the dropdown
- Click **Update IAM Role**

> ✅ Your instance now assumes the IAM role with access to S3 (or other services).

✅ Part 3: Install AWS CLI on EC2

1. SSH into your EC2

```bash
ssh -i your-key.pem ubuntu@your-ec2-public-ip
```

2. Install AWS CLI (for Ubuntu/Debian)

```bash
sudo apt update
sudo apt install awscli -y
```

Verify:

```bash
aws --version
```

> ❗ No need to run `aws configure` — the CLI will automatically use the **EC2 instance role** credentials.

✅ Part 4: Use AWS CLI to Access Resources

Example: List S3 Buckets

```bash
aws s3 ls
```

Example: Copy a file to an S3 bucket

```bash
echo "Hello from EC2" > hello.txt
aws s3 cp hello.txt s3://your-bucket-name/
```

🛡️ Security Notes

- Always use **least privilege** when assigning permissions to roles
- Avoid `FullAccess` in production if not necessary
- For auditing, use CloudTrail to monitor actions taken via IAM roles

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✅ Summary

| Step | Task |

1	Created IAM role with S3 access via GUI
2	Attached IAM role to EC2 instance
3	Installed AWS CLI on EC2
4	Used CLI to interact with AWS securely
